batmonitor 0.56-c: one cycle every 5 s | blue = decision (bits + state, make check) | green = matrix / amount | details: charge_flow.pdf, discharge_flow.pdf

inputs (bm_parse: only valid numbers, else stale)
PCC, Sofar Bat1, PV, heat pump W, R290 Hz, BMS SoC / U / CCL / DCL / lowest cell, season, forecast

D1 forecast, B1FIRST, WP_LOW (0.54-c)
-> S2 heat pump kind NONE / CAP / ALL / BAD (z2 = PCC + WP not covered)
-> S3 source DAY / B1 / STK / B1W
make check: 442368 combinations (16 bits x 3 seasons x 8 states)

D5 PCC > 20 kW for 2 min (at once > 23 kW) -> DO4 pulse
pv_relay/DO4 (FoxESS shedding)
first: R290 boost + air conditioning at ≥ 18 kW
grid offline: AC-out feed > 5 kW / unit for 10 s -> at once

S0 caps per stack: chg_cap, dis_cap (charger, BMS CCL / DCL)

heat pump matrix (S2 kind from D1, + D6 WPPV): who pays the heat pump

WP reason	when	S2 kind	discharge: stacks cover	charge point z2 / chg_ref	grid buys
OFF / NA	WP < 300 W / em0 stale	-	house only (NA + winter: WP_CAP 1 kW)	PCC	-
LOWSOC	lowest stack < WP_SOC_MIN 21 % (back ≥ 23 %)	NONE	0 W of the WP	PCC + WP	all of the WP
Z2	winter	NONE	0 W of the WP	PCC + WP	all of the WP
T1900 / ALL	transition	CAP	up to 1900 W	PCC + (WP - 1900)	WP above 1900 W
FCBAD	bad forecast (target_soc 100)	BAD	50 % of the WP	PCC + 50 % WP	50 % of the WP
FC / ALL	good forecast / summer	ALL	all of the WP	PCC	-
... + WPPV (0.56-c)	kind ALL, day, WP on, PV covers house + WP, no D4 cap	ALL	(charging, no discharge)	PCC + min(50 % WP, 2000 W)	min(50 % WP, 2 kW) into the stacks

charge path

D4 forecast export cap (May-Aug)
while the forecast expects export > 18 kW:
wait below the cap, charge only above it

D6 WPPV (0.56-c)
WP on PV, kind ALL, day, no D4 cap:
chg_ref += min(WP_PV_CHG_SHARE 50 % x WP, WP_PV_GRID_MAX 2 kW)
on surplus > 200 W, held while ≥ 0
make check: 384 combinations

D2 mode per stack, first match:
BMS_MISSING > FORCE > STALE > LADESPERRE > CHARGE / FULL > PROT > CHARGING > DISFORCE = SoC < SOC_FORCE or LOW (DCL 0 / cell < 3.05 V; ends at SOC_MIN_RELEASE, 0.55-c)
make check: 16128 combinations per stack (84 bit sets x 12 global x 16 states) -> mode:
BMS_MISSING 8064 · FORCE 5760 · STALE 1152 · LADESPERRE 288 · CHARGE 216 ·
PROT 216 · CHARGING 180 · DIS 180 · FULL 72

D-lead: stack more than SOC_BALANCE_ON ahead

mode matrix BM_MODE_RULE
0 W | FORCE | charge | discharge

charge amount
max(0, own + KP x (z2 + Bat1_eff - target))
chg_ref = z2 (+ D6 share); target + EXPORT_CAP while D4 cap;
≤ sum of chg_cap

FORCE: min(500 W, chg_cap)
any time of day, no TOU window

alloc_levels up to chg_cap
stack by stack in CHARGE_PRIORITY,
lead stack last

discharge branch

BM_SRC_RULE[src]
STK: Bat1 signed, +trickle, idle 0
else: Bat1 charge only, idle 20 W

D3 PROP / IDLE

discharge amount
min(PROP ? deficit : idle, W_MAX, WP_CAP)

alloc_levels up to dis_cap
lead stack first, else
every stack the same share

setpoints per phase -> Multis (+ charge / - discharge)
rule text: mode WP: SRC: split [CAP] [WPPV] [CCL / DCL / WPCAP]
-> state.json -> victron2db.py -> wagodb.pv_victron.bm_l1..3_rule (every minute)

no cap
kind ALL
S2 kind
block
surplus
SoC
src, z2
+ share
CHARGE
FORCE
DIS
0 W rows